48 Keys Keyboard: Keymap

QMK Vial firmware for the RP2040 (Raspberry Pi Pico) controller. Used by Micro Journal Rev.2 2.0 Revamp, Rev.2 2.1 Mini and Rev.3 3.0 Revamp.

How layers work

There are only 48 keys, so each key can do more than one thing. The firmware has four **layers**. A layer is a full set of key assignments, and the keyboard switches layers while you hold a layer key.

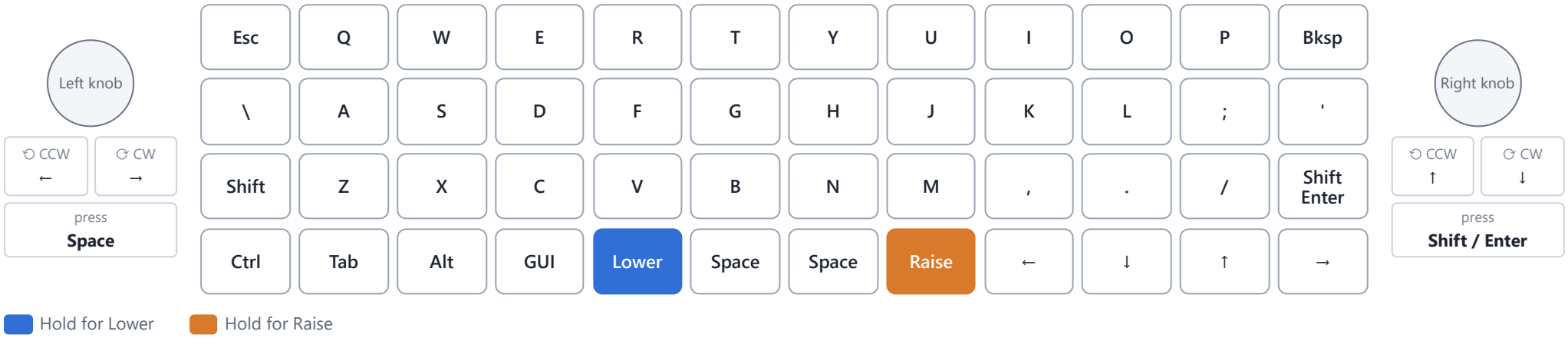
- Base is active when no layer key is held: letters, Space, Enter and the arrow keys.
- Hold Lower to get Lower: the shifted symbols ! @ # $..., Delete, Home/End.
- Hold Raise to get Raise: numbers, F1 to F12, and - = [ ].
- Hold Lower and Raise together to get Nav: Home/End and Page Up/Down on the knobs, and the bootloader key.

Release the layer key and you're back on Base. On a layer, a key that has nothing assigned (dashed in the diagrams) does what it does on Base.

Special keys

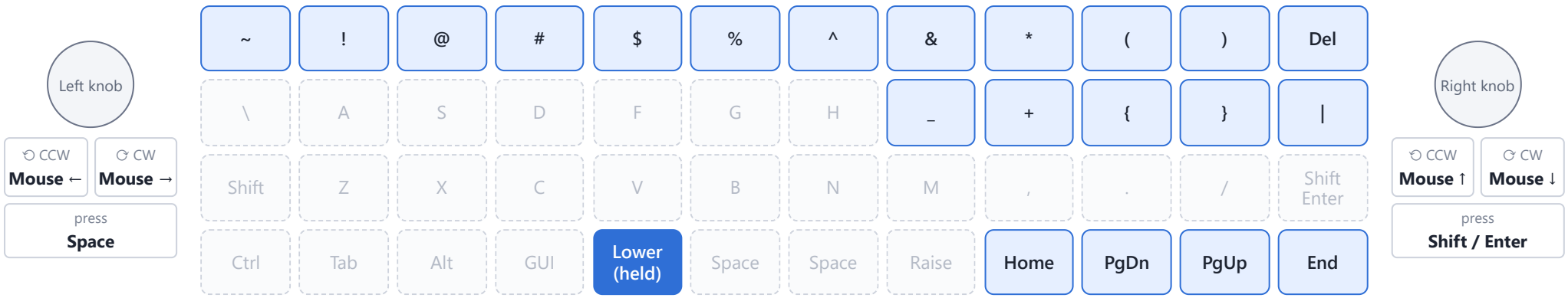
Key	What it does
Shift / Enter	Tap for **Enter**, hold for **Right Shift**.
Lower Raise	Hold one for its layer. Hold both for Nav.
Knobs	Turning a knob sends a key, and the key changes with the layer (see the knob table on page 3). Pressing the left knob is **Space**, pressing the right knob is **Shift / Enter**.
Boot	On Nav (Lower + Raise + Q): restarts the Pico as the `RPI-RP2` drive so you can flash new firmware.

Base Default layer



Lower Hold Lower: symbols

The shifted symbols, so you can type them without Shift. Backspace becomes Delete, and the arrow keys become Home / Page Down / Page Up / End.



Raise Hold Raise: numbers and function keys

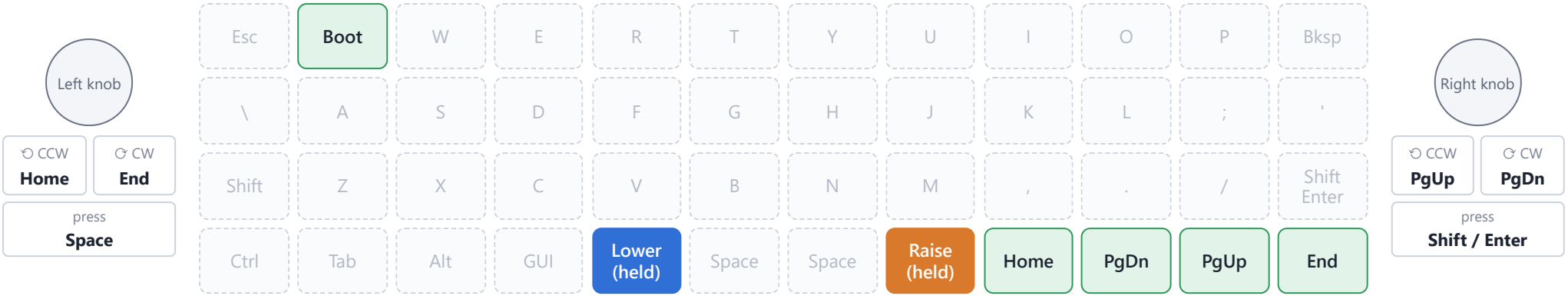
The number row, F1 to F12 on the left half, and `- = [ ]` on the right. The arrow keys become Home / Page Down / Page Up / End.



Changed on this layer Not assigned: works as on Base

Nav Hold Lower + Raise

For moving around long documents: the knobs jump by line start/end and by page. Boot on the Q key is for flashing firmware.



Knobs on each layer

Layer	Left knob counter-clockwise / clockwise	Right knob counter-clockwise / clockwise
Base	Left arrow / Right arrow	Up arrow / Down arrow
Lower	Mouse left / Mouse right	Mouse up / Mouse down
Raise	Scroll up / Scroll down	Scroll left / Scroll right
Nav	Home / End	Page Up / Page Down

Pressing a knob sends the same key on every layer: left knob = Space, right knob = Shift / Enter.

Changing the keymap with Vial

- Open **get.vial.today** in Chrome or Edge, or use the Vial desktop app, and connect the keyboard over USB.
- Click a key, then pick a new keycode. Changes are saved on the keyboard right away. No flashing needed.
- The knobs can be reassigned too, separately for each layer.
- To unlock Vial (it asks before some changes), hold Esc + Q when it tells you to.

Entering the bootloader

- Press Lower + Raise + Q (Boot), or
- hold Esc while plugging in the USB cable, or
- hold the **BOOTSEL** button on the Pico while plugging it in.

The Pico shows up as the `RPI-RP2` drive. Copy `unkyulee_vial.uf2` onto it.

Source: shared/keyboard-48/rp2040/qmk-vial/keymaps/vial/keymap.c. This is the factory default; anything changed in Vial overrides it.